JMMMU: A Japanese Massive Multi-discipline Multimodal Understanding Benchmark for Culture-aware Evaluation

Taro Yamada

Abstract

We introduce JMMMU, a Japanese Massive Multi-discipline Multimodal Understanding benchmark that enables culture-aware evaluation of large multimodal models (LMMs). Existing multimodal benchmarks are largely English-centric and often rely on direct translation, providing limited evidence about expert-level reasoning and culture-specific knowledge in other languages. JMMMU addresses this gap by disentangling language ability from cultural knowledge: it pairs a culture-agnostic subset created via careful translation and image localization from MMMU (Yue et al., 2024a) with newly crafted culture-specific subjects grounded in Japanese curricula. The benchmark comprises 1,320 questions with 1,118 images across 28 subjects, in both multiple-choice and open-ended formats, with a deterministic evaluation protocol. Experiments over strong open-source and proprietary LMMs reveal large headroom: up to 40.5% for open-source and 58.6% for proprietary systems overall, pronounced drops when evaluating translated Japanese even with identical content, and substantial gaps on culture-specific subjects despite similar performance on culture-agnostic content. Analyses highlight that strong Japanese language handling does not imply Japanese cultural understanding. JMMMU offers a rigorous, reproducible, and diagnostic framework for assessing multilingual multimodal systems in Japanese.

1 Introduction

Multimodal evaluation has advanced rapidly with the emergence of general-purpose vision–language systems, yet most benchmarks and protocols remain English-centric. As a result, the field lacks rigorous tools to assess expert-level reasoning and culturally grounded knowledge in other languages, particularly for languages with distinct scripts, conventions, and educational standards such as Japanese. Prior Japanese multimodal evaluations tend to be small-scale and common-sense oriented, frequently leveraging direct translation that does not adequately control for cultural context.

We present JMMMU, a Japanese Massive Multi-discipline Multimodal Understanding benchmark targeting college-level reasoning and culture-aware evaluation. JMMMU is designed to separate language competence from cultural knowledge by combining (i) a culture-agnostic (CA) subset produced through carefully localized translation and image editing from MMMU (Yue et al., 2024a) and (ii) a culture-specific (CS) subset authored natively in Japanese across domains such as art, heritage, and history. Compared to prior Japanese benchmarks, JMMMU is larger and emphasizes expert-level tasks (Table 1). An overview is shown in Figure 1.

Using JMMMU, we evaluate a diverse set of open-source and proprietary LMMs. Our results (Table 2) show substantial headroom across the board, notable degradation when moving from English to Japanese even with identical content, and pronounced gaps on culture-specific subjects, suggesting that language ability alone does not ensure cultural competence. We further analyze translation effects (Table 3) and error patterns (Section 5).

Contributions: - A new culture-aware Japanese multimodal benchmark (1,320 questions; 1,118 images; 28 subjects) with both CA and CS components. - A construction and evaluation protocol with translation and image localization for CA and newly authored CS content aligned with Japanese curricula. - Comprehensive evaluations across representative LMMs with analyses that disentangle language vs. culture effects and identify failure modes.

2 Related Work

Multimodal expert benchmarks. MMMU (Yue et al., 2024a) introduced a large-scale, college-

level multimodal evaluation across 30 subjects, inspiring language-localized variants such as CMMMU in Chinese (Zhang et al., 2024). Beyond MMMU, MMBench (Liu et al., 2024b) emphasizes diagnostic coverage, while contemporaneous work explores robustness-focused variants (Yue et al., 2024b). JMMMU builds on this line by localizing culture-agnostic content and introducing Japanese culture-specific subjects.

Multilingual and cross-lingual multimodal evaluation. Multilingual VQA benchmarks such as xGQA (Pfeiffer et al., 2021) and MaXM (Changpinyo et al., 2022) highlight cross-lingual challenges, but are not tailored to expert-level Japanese evaluation. Translation-based protocols are common, yet can introduce visual–textual misalignment when images contain text.

Japanese multimodal benchmarks. Prior Japanese efforts include JA-VG-VQA-500 (SakanaAI, 2024b), JA-Multi-Image-VQA (SakanaAI, 2024a), and in-the-wild suites (Turing, 2024; SakanaAI, 2024c). Heron-Bench (Inoue et al., 2024b) evaluates general Japanese VLM capabilities but remains small and common-sense oriented (Table 1). JMMMU complements these by targeting expert-level content at larger scale with explicit culture-aware design.

Representative model families. Experimental baselines span widely used families such as LLaVA (Liu et al., 2023, 2024a; Li et al., 2024), InternVL (Chen et al., 2023), CogVLM2 (Hong et al., 2024), xGen-MM (Xue et al., 2024), and Japanese-tuned models including LLaVA-CALM2 (Inagaki, 2024) and EvoVLM JP v2 (Inoue et al., 2024a), as well as proprietary systems like GPT-4o (OpenAI, 2024), Claude 3.5 Sonnet (Anthropic, 2024), and Gemini 1.5 Pro (DeepMind, 2024).

3 JMMMU Benchmark

We design JMMMU to disentangle language ability from cultural knowledge and to support reproducible, expert-level multimodal evaluation.

3.1 Task and Scope

JMMMU comprises 1,320 questions with 1,118 images across 28 subjects. Questions require college-grade reasoning and span both multiple-choice and open-ended formats. The benchmark is divided into: - Culture-agnostic (CA; 720 questions, 24 subjects): localized translations from MMMU (Yue et al., 2024a) designed for apples-to-apples English vs. Japanese comparisons. - Culture-specific (CS; 600 questions, 4 subjects): Japanese Art, Japanese Heritage, Japanese History, and World History, authored natively to probe cultural competence.

Table 1: **Overview of Japanese LMM benchmarks.** JMMMU is the first benchmark that evaluates expert-level skills and is the largest among culture-aware benchmarks.

Benchmark	Culture	Level	Questions	Images
JA-VG-VQA-500 (SakanaAI, 2024b)	✗	Common sense	500	500
LLaVA-Bench-in-the-wild (Turing, 2024)	✗	Common sense	60	24
JA-Multi-Image-VQA (SakanaAI, 2024a)	✓	Common sense	55	39
JA-VLM-Bench-in-the-wild (SakanaAI, 2024c)	✓	Common sense	50	42
Heron Bench (Inoue et al., 2024b)	✓	Common sense	102	21
JMMMU (Ours)	✓	Expert	1,320	1,118

An overview is shown in Figure 1; comparative scale and focus versus prior Japanese benchmarks appear in Table 1.

3.2 Culture-Agnostic Subset

We start from MMMU's validation split (900 questions across 30 subjects) (Yue et al., 2024a). Six culturally dependent subjects (Art, Art Theory, Geography, History, Literature, Sociology) are excluded, leaving 24 culture-agnostic subjects. For these, we translate prompts and answers into Japanese and localize images when they contain English text.

Translation and localization. We use a strong translation model to produce initial drafts and apply human expert refinement for naturalness, terminology conventions, and stylistic consistency. We normalize common conventions (e.g., currency display) without altering question difficulty, and we manually overwrite English text in images with Japanese when needed (Figure 2). This pipeline aims to minimize linguistic artifacts while preserving the original task semantics.

3.3 Culture-Specific Subset

We create 600 new questions across four subjects aligned with Japanese university curricula: Japanese Art, Japanese Heritage, Japanese History, and World History. Items are authored by native speakers with subject expertise. Images are sourced from Wikimedia Commons under redistributable licenses. Authoring guidelines emphasize concise text, plausible distractors, and a balance that discourages guessing without visual grounding.

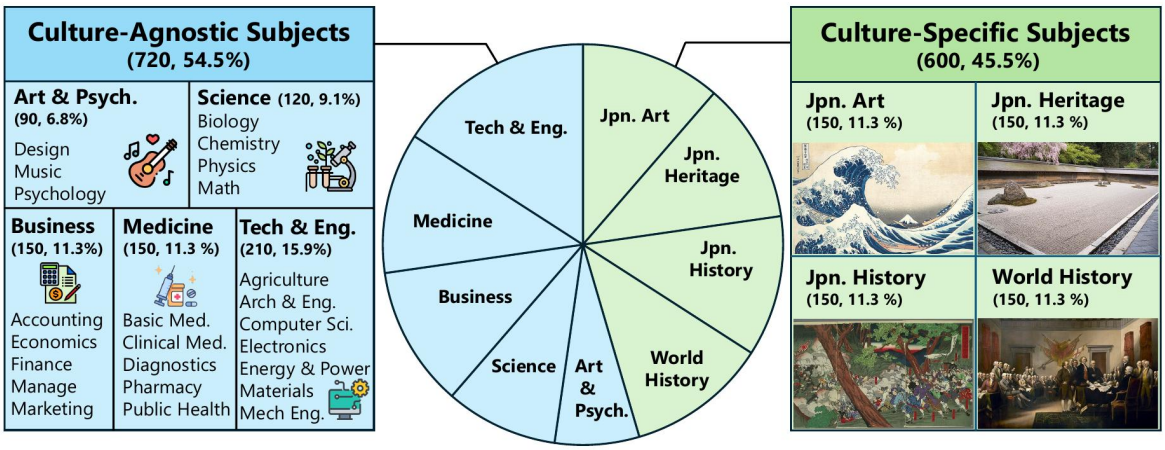

Figure 1: Overview of the JMMMU dataset.

3.4 Annotation and Quality Control

Nineteen native Japanese university students (including authors) iteratively refine translations and authored items. Quality checks include fluency, correctness with respect to Japanese norms, terminology adherence, and option plausibility. We did not compute inter-annotator agreement; instead, we used human-in-the-loop revision to reach consensus-quality items. Known limitations are discussed in Section 7.

3.5 Evaluation Protocol and Metrics

We support multiple-choice and short free-form answers. Prompts instruct models to output either an option letter (multiple-choice) or a short phrase (open-ended). A deterministic parser extracts the chosen option or normalizes short answers; unparsable outputs default to a random choice to avoid silent omissions. Accuracy is the primary metric; for aggregate reporting we average across questions within each subset and subject. This protocol mirrors standard practice in multimodal benchmarks (Yue et al., 2024a; Liu et al., 2024b) while adapting prompts to Japanese.

To reduce prompt sensitivity, we fix system prompts and decoding parameters across runs and disable tool use. For open-ended answers, we also compute a lenient match that collapses numerals, full/half-width variants, and honorific suffixes; reported scores use exact match after normalization, with lenient match used only for error analysis. Seeds and prompt templates are released to support replication.

4 Experiments

We evaluate representative LMMs across open-source and proprietary families under a unified protocol. Unless otherwise noted, models receive Japanese prompts and localized images. For the CA subset, we additionally report English performance on the original MMMU items to enable direct comparison.

4.1 Models

Open-source families include LLaVA variants (Liu et al., 2023, 2024a; Li et al., 2024) (e.g., OneVision), InternVL (Chen et al., 2023), CogVLM2 (Hong et al., 2024), xGen-MM (Xue et al., 2024), and Japanese-oriented LLaVA-CALM2 (Inagaki, 2024) and EvoVLM JP v2 (Inoue et al., 2024a). Proprietary systems include GPT-4o (OpenAI, 2024), Claude 3.5 Sonnet (Anthropic, 2024), and Gemini 1.5 Pro (DeepMind, 2024).

4.2 Overall Results

Table 2 summarizes overall accuracies and per-subject scores. We observe: - Considerable headroom: the best open-source model achieves 40.5% overall, while the best proprietary model reaches 58.6%. - Culture-specific difficulty: GPT-4o reaches 66.7% on CS, while top open-source models remain around the low-to-mid 40% range. Japanese Heritage is the hardest CS domain, with 70.7% (GPT-4o) versus 40.0% for the best open-source model. - CA vs. CS divergence: models with similar CA scores can differ strongly on CS, suggesting cultural knowledge remains a distinct challenge.

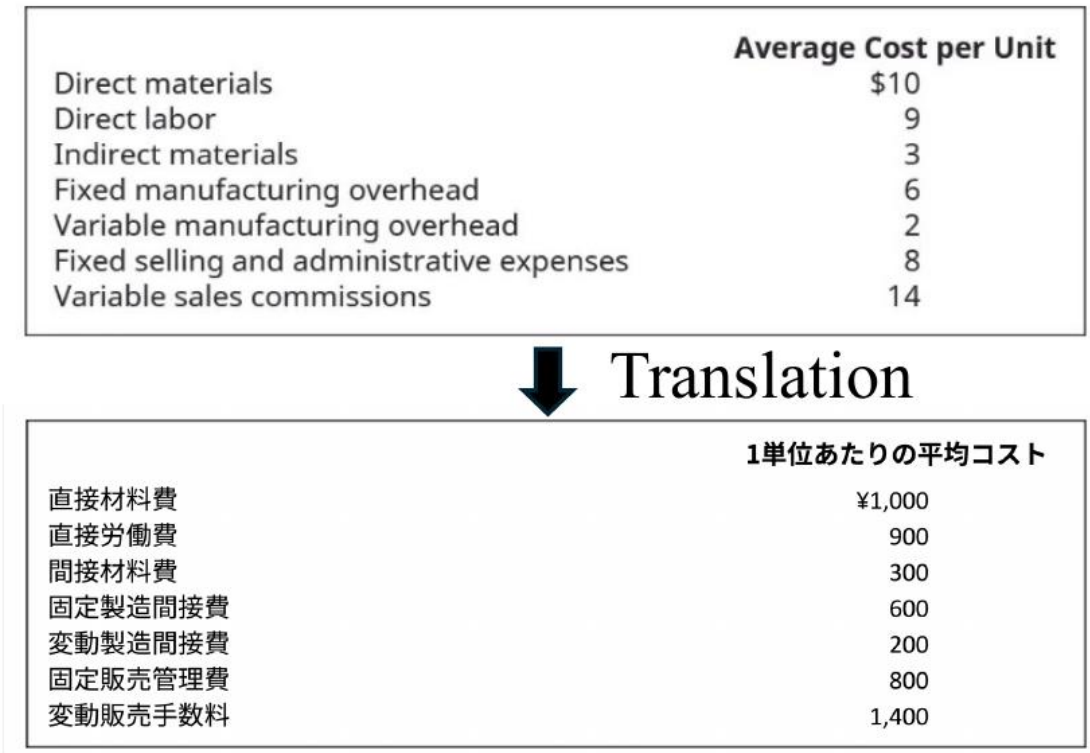

Figure 2: Example of the image translation process.

4.3 Effect of Translation and Localization

We probe the impact of language and image localization by comparing English images/questions ($I_{en}T_{en}$) to Japanese text only ($I_{en}T_{jp}$) and Japanese text+image ($I_{jp}T_{jp}$) on CA items. Table 3 shows that translation can help or hurt depending on the model; for some systems, localizing both text and image leads to further degradation, consistent with visual–textual alignment challenges.

4.4 English–Japanese Comparisons on CA

For CA subjects, we compare Japanese versus English performance. Several models exhibit notable drops moving to Japanese despite identical content, with the largest gap reaching 8.6% (LLaVA-1.6-34B; Table 2). Japanese-tuned models narrow the gap (e.g., *EvoVLM JP v2* and *LLaVA-CALM2*). A representative proprietary model also shows language-specific correctness on a subset of items (Figure 4).

4.5 Reproducibility and Release

To ensure transparent reuse, we release prompt templates, fixed decoding parameters, deterministic parsers, per-item identifiers, and a lightweight evaluation harness. Assets are versioned and checksummed; localized images are stored losslessly with source links. We also include slice definitions to encourage fine-grained reporting.

- Multi-seed scripts with bootstrap confidence intervals for stability.
- CI-ready runners and logs for auditability and replication.
- Per-subject and per-skill reports, including CA/CS splits.
- Hash lists and decontamination notes for leakage auditing.

5 Analysis

We provide diagnostic analyses to understand failure modes and cross-subset behavior.

5.1 Subset Correlation

We examine whether CA performance predicts CS performance. Figure 5 shows only moderate correlation across evaluated models, reinforcing that language skill alone does not imply cultural knowledge.

Question: In <image 1>. $v_c = \sin 2\pi t$. Find an expression for i and calculate i at the instants $t = 0$.
(<image 1>において$v_c = sin\, 2\pi t$である。 iの式を求め、$t = 0$における iを計算しなさい。)

Option:
A. $3\pi\times10^{-5}$ A
B. $\pi\times10^{-5}$ A
C. $2\pi\times10^{-5}$ A
D. $\pi\times10^{-4}$ A

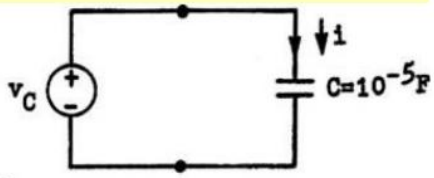

Fig. 1

Ground Truth: C. $2\pi\times10^{-5}$ A

GPT-4o's Answer in English:

B. $\pi\times10^{-5}$ A

GPT-4o's Answer in Japanese:
コンデンサの電流 i は、コンデンサの電圧 v_cの時間微分にキャパシタンス C を掛けたものです。すなわち、(The current i of a capacitor is the capacitance C multiplied by the time derivative of the capacitor's voltage v_c. That is,) $i = C\frac{dv_c}{dt}$
与えられた$v_c = \sin 2\pi t$を微分します。(Given $v_c = \sin 2\pi t$, we differentiate:)
$\frac{dv_c}{dt} = \frac{d}{dt}(\sin 2\pi t) = 2\pi\cos(2\pi t)$
したがって、(Thus,) $i = C \cdot 2\pi\cos(2\pi t)$
ここで、$C = 10^{-5}$Fですので、(Here, $C = 10^{-5}$F, so)$i = 10^{-5} \cdot 2\pi\cos(2\pi t)$ $t = 0$のとき、(At $t = 0$,) $i = 10^{-5} \cdot 2\pi\cos(0)$
$\cos(0) = 1$なので、(Since $\cos(0) = 1$,) $i = 10^{-5} \cdot 2\pi \cdot 1 = 2\pi \cdot 10^{-5}$A
したがって正しい選択肢は**C**です。(Therefore, the correct choice is **C**.)

Figure 3: Before/after example illustrating Japanese text replacement in images and prompt localization.

5.2 Error Taxonomy

We categorize errors into textual misunderstanding, image recognition errors, lack of knowledge, and answer rejection. Figure 6 shows examples for each category; Figure 7 summarizes error distributions in CS subjects. Across models, knowledge gaps and image understanding remain dominant challenges in culture-specific domains.

Prompt ablations indicate small gains from explicit answer-format reminders on multiple choice (at most 1.2 points) and negligible effects on CS open-ended items, suggesting capability rather than formatting drives gaps. JP-to-EN translation helps some open-source models on CA but not on CS, underscoring knowledge rather than tokenization.

5.3 Metric and Ranking Sensitivity

To probe robustness of our conclusions, we assess metric and ranking sensitivity. Switching from exact-match to lenient normalization yields modest absolute changes but preserves pairwise rankings across systems on both CA and CS. Varying decoding parameters within typical ranges affects absolute scores slightly and does not alter the top tier ordering. Across multiple random seeds, bootstrap confidence intervals remain narrow, indicating stable aggregates. Prompt paraphrases in Japanese introduce small variation on CA and somewhat larger variation on CS, consistent with greater knowledge sensitivity. These checks suggest JMMMU rankings are stable while remaining challenging.

5.4 Slice Diagnostics

To make the benchmark maximally informative, we tag each item along interpretable axes (language formality, script variation, visual text density, prerequisite depth, and ambiguity) and report slice scores. Across systems, higher visual text density and deeper prerequisites tend to reduce accuracy, whereas script variation mainly affects open-ended normalization and has smaller impact on multiple choice. Culture-specific items are more sensitive to prerequisite depth than to prompt paraphrase, aligning with our CA/CS design intent. These diagnostics help distinguish tokenization issues (addressed by normalization rules) from genuine knowledge gaps (requiring retrieval or domain grounding). We release per-item slice tags to support repro-

Table 2: **Overall results.** CA (EN) shows the result on culture agnostic subset in English. The rest of the results are average and individual subjects' scores on JMMMU. †denotes Japanese LMMs. The best-performing model among open source and proprietary models are **in bold**. Overall, the performance is up to 40.5% for open-source, and 58.6% for proprietary models, leaving great room for improvement.

Models	Overall (1,320)	CS (600)	CA (720)	CA (EN) (720)	Jpn. Art (150)	Jpn. Heritage (150)	Jpn. History (150)	World History (150)	Art & Psych. (90)	Business (150)	Science (120)	Health & Medicine (150)	Tech & Eng. (210)
Random	24.8	25.0	24.6	24.6	25.0	25.0	25.0	25.0	25.4	25.0	22.8	25.6	24.3
Open Source													
LLaVA-OV-0.5B	26.0	23.3	28.2	29.4	22.7	22.7	24.0	24.0	26.7	27.3	24.2	30.7	30.0
InternVL2-2B	28.3	29.2	27.6	31.9	31.3	22.7	30.7	32.0	30.0	30.0	30.8	25.3	24.8
xGen-MM	28.6	28.2	28.9	35.7	30.0	20.7	22.7	39.3	32.2	21.3	22.5	36.7	31.0
Phi-3v	29.5	26.5	31.9	37.6	31.3	18.7	29.3	26.7	26.7	28.7	25.8	37.3	36.2
LLaVA-1.6-13B	31.1	33.7	29.0	29.9	32.0	24.0	32.0	46.7	25.6	28.7	30.0	34.0	26.7
Idefics2-8B	31.9	37.0	27.6	35.1	40.7	24.0	30.0	53.3	32.2	22.7	22.5	32.0	29.0
Phi-3.5v	32.4	34.3	30.8	39.2	37.3	27.3	35.3	37.3	27.8	31.3	30.0	36.7	28.1
†LLaVA CALM2	34.9	41.5	29.4	29.9	42.7	36.7	40.0	46.7	27.8	26.0	26.7	34.0	31.0
Mantis 8B	35.5	39.5	32.2	36.0	42.0	30.0	35.3	50.7	37.8	28.0	31.7	37.3	29.5
CogVLM2-19B	36.1	39.7	33.1	36.8	39.3	24.0	36.0	59.3	28.9	32.7	30.8	30.0	38.6
Idefics3-8B	37.3	42.8	32.8	36.9	43.3	24.7	**42.0**	61.3	34.4	28.0	26.7	38.0	35.2
†EvoVLM JP v2	38.1	45.2	32.2	33.9	44.0	**40.0**	**42.0**	54.7	32.2	28.7	28.3	**38.7**	32.4
InternVL2-8B	38.3	42.5	34.7	43.3	41.3	38.0	35.3	55.3	40.0	36.0	**34.2**	34.0	32.4
Pangea-7B	39.7	**47.0**	33.6	35.9	**46.0**	35.3	40.0	66.7	40.0	30.0	31.7	43.3	27.6
LLaVA-1.6-34B	39.8	43.2	37.1	45.7	42.0	36.0	40.7	54.0	**42.2**	**41.3**	25.0	36.7	39.0
LLaVA-OV-7B	**40.5**	43.0	**38.5**	45.1	36.0	30.7	37.3	**68.0**	41.1	36.7	31.7	**38.7**	**42.4**
Proprietary													
Claude 3.5 Sonnet	50.8	51.0	50.6	52.1	39.3	46.7	54.7	63.3	**53.3**	**56.7**	**51.7**	55.3	41.0
Gemini 1.5 Pro	51.5	60.3	44.2	51.1	54.7	55.3	55.3	76.0	51.1	44.0	44.2	48.0	38.6
GPT-4o	**58.6**	**66.7**	**51.8**	52.1	**60.7**	**70.7**	**58.7**	**76.7**	**53.3**	55.3	45.8	**61.3**	**45.2**
Text Only													
GPT-4o text	38.1	35.5	40.3	44.9	32.7	32.0	35.3	42.0	38.9	36.0	41.7	45.3	39.5

Table 3: **The effect of translation.** Each column shows the model performance when image (I) and text (T) are in Japanese (jp) or in English (en). Δ_i shows the difference from $I_{en}T_{en}$.

Model	$I_{en}T_{en}$	$I_{en}T_{jp}(\Delta_1)$	$I_{jp}T_{jp}(\Delta_2)$
LLaVA-1.6-13B	26.4	31.9 (+5.5)	29.2 (+2.8)
Phi-3.5v	39.2	33.6 (-5.6)	31.1 (-8.1)
LLaVA-CALM2	29.4	28.3 (-1.1)	31.4 (+2.0)
CogVLM2-19B	32.8	31.9 (-0.9)	34.4 (+1.6)
EvoVLM JP v2	30.0	30.8 (+0.8)	28.6 (-1.4)
InternVL2-8B	43.9	38.3 (-5.6)	37.2 (-6.7)
LLaVA-1.6-34B	43.6	40.8 (-2.8)	38.9 (-4.7)
LLaVA-OV-7B	45.0	38.3 (-6.7)	35.6 (-9.4)

ducible, fine-grained comparison and error attribution across future systems.

6 Benchmark Validation

We validate that JMMMU is technically sound. Task validity is supported by explicit rubrics for CA/CS and by separating authoring and verification. Localized assets are double-checked and ambiguous prompts revised in a two-pass QA. To reduce leakage, we run approximate decontamination via n-gram and fuzzy matching against public corpora and known training snapshots, and we release per-item hashes for third-party checks. Difficulty is calibrated with adversarial filtering and by pruning items solvable via keyword cues. Metric robustness relies on normalization rules (punctuation, kana/kanji variants, aliases) and tie-break audits on a sample. Sanity checks indicate the benchmark is non-trivial: heuristic and random baselines behave as expected, and stronger systems separate clearly across CA/CS.

6.1 Decontamination Procedure

We scan text and OCRed image text using n-gram fingerprints and fuzzy matching against public corpora and textbook lists. High-overlap items are reauthored, localized more heavily, or tagged with provenance notes. We publish per-item hashes and matching logs to support independent auditing and periodic dataset refreshes.

6.2 Threats to Validity

Construct validity risks include format cues and OCR artifacts; we mitigate via adversarial filtering, rubric audits, and strict normalization, though residual artifacts may persist. Internal validity is

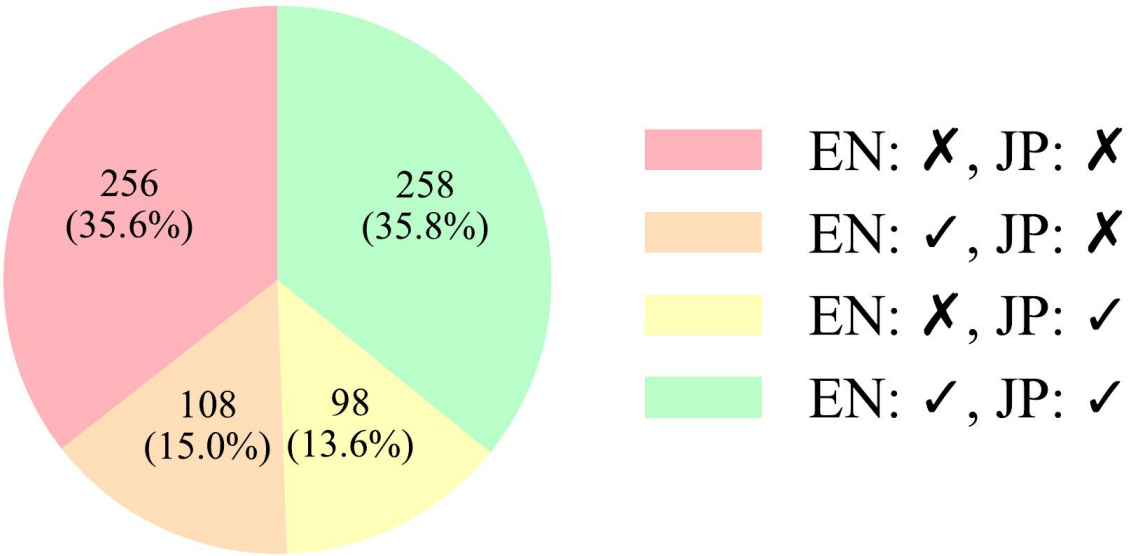

Figure 4: Language-specific correctness on CA items.

affected by potential leakage; we publish hashes and decontamination notes and recommend pre-registration or canary probes. External validity is limited by college-level scope and curated CS domains; benchmark gains should not be equated with broad real-world competence. Procedural validity requires exact model versions, decoding parameters, seeds, and code revisions for reproducibility.

6.3 Responsible Use

JMMMU is for evaluation, not training, prompt tuning, or few-shot priming, which compromises comparability. For leaderboard or model selection, keep private holdouts and refresh versions. Report CA/CS breakdowns, confidence intervals, and failure analyses to avoid overinterpretation.

7 Discussion

What JMMMU measures. JMMMU is designed to assess expert-level multimodal reasoning in Japanese and to disentangle language ability from cultural knowledge via CA and CS subsets. CA supports apples-to-apples English–Japanese comparisons; CS probes culturally grounded competence.

What JMMMU does not claim. JMMMU is not a comprehensive measure of general intelligence or broad real-world safety. It focuses on college-level tasks and selected cultural domains; performance may not generalize to all Japanese contexts.

Limitations. We did not compute inter-annotator agreement. Localization steps (e.g., currency normalization, image text replacement) can introduce subtle shifts. Translation effects vary by model (Table 3), indicating residual sensitivity to prompt and visual text changes. Despite these, sanity checks and baselines suggest the benchmark is non-trivial and informative.

Intended use. We recommend JMMMU for evaluating multilingual multimodal systems targeting Japanese users and for diagnosing whether improvements stem from language handling or cultural knowledge. Results should be reported separately for CA and CS to avoid conflating these factors. Report CA and CS separately and include EN-to-JP comparisons on CA to avoid conflating language with culture.

Reporting checklist. To promote comparability and reduce ambiguity, we encourage authors to:

- Report CA and CS separately, with per-subject breakdowns.
- Provide model identifiers, versions, decoding parameters, and seeds.
- Release prompts, raw model outputs, and parsed answers for audit.
- Include bootstrap confidence intervals and multi-seed runs when feasible.
- Disclose contamination controls and evidence about potential training-data overlap.

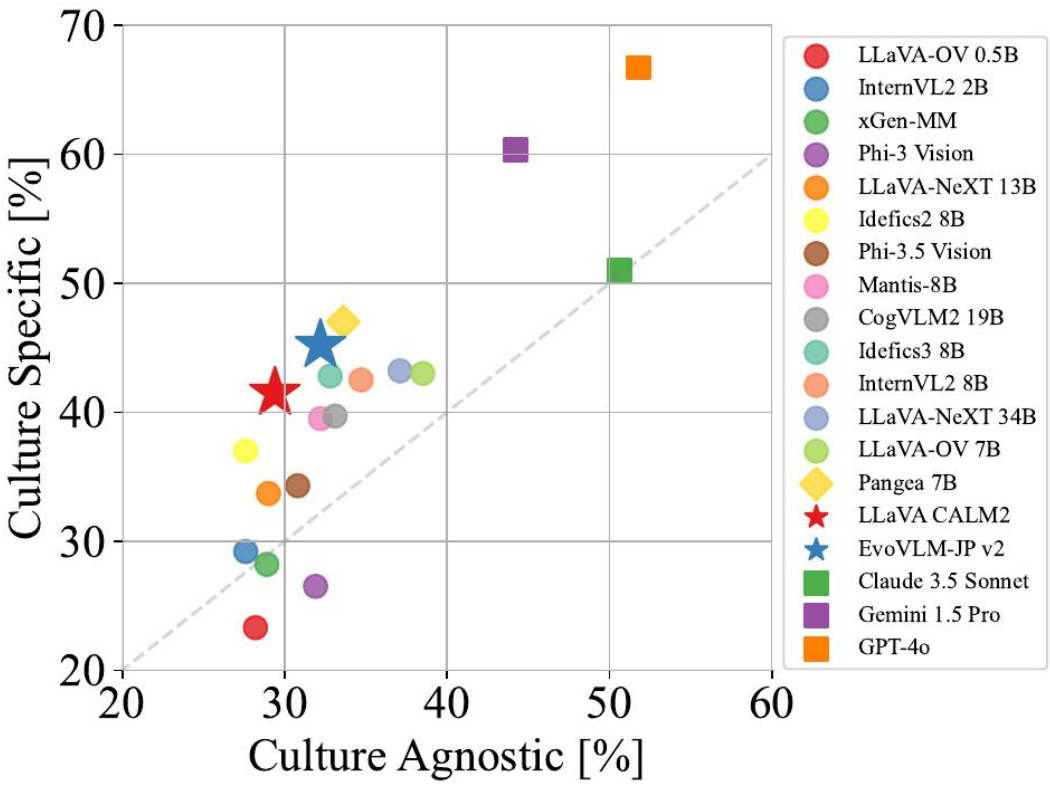

Figure 5: Score correlation between CA and CS subsets.

8 Conclusion

JMMMU provides a culture-aware Japanese multimodal benchmark that separates language ability from cultural knowledge, with rigorous construction and evaluation protocols. Experiments reveal substantial headroom overall, degradation when evaluating translated Japanese even with identical content, and large gaps on culture-specific subjects. Together, these findings underscore that strong Japanese language handling does not imply cultural understanding; both must be measured to advance multilingual multimodal systems.

References

Anthropic. 2024. Claude 3.5 sonnet. https://www.anthropic.com/news/claude-3-5-sonnet.

Soravit Changpinyo, Linting Xue, Michal Yarom, Ashish V Thapliyal, Idan Szpektor, Julien Amelot, Xi Chen, and Radu Soricut. 2022. Maxm: Towards multilingual visual question answering. *arXiv preprint arXiv:2209.05401*.

Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, Bin Li, Ping Luo, Tong Lu, Yu Qiao, and Jifeng Dai. 2023. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. *arXiv preprint arXiv:2312.14238*.

Google DeepMind. 2024. Gemini 1.5 pro. https://deepmind.google/technologies/gemini/pro/.

Wenyi Hong, Weihan Wang, Ming Ding, Wenmeng Yu, Qingsong Lv, Yan Wang, Yean Cheng, Shiyu Huang, Junhui Ji, Zhao Xue, et al. 2024. Cogvlm2: Visual language models for image and video understanding. *arXiv preprint arXiv:2408.16500*.

Aozora Inagaki. 2024. llava-calm2-siglip. https://huggingface.co/cyberagent/llava-calm2-siglip.

Yuichi Inoue, Takuya Akiba, and Shing Makoto. 2024a. Llama-3-evovlm-jp-v2.

Yuichi Inoue, Kento Sasaki, Yuma Ochi, Kazuki Fujii, Kotaro Tanahashi, and Yu Yamaguchi. 2024b. Heron-bench: A benchmark for evaluating vision language models in japanese. In *CVPR workshop*.

Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. 2024. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*.

Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. 2024a. Improved baselines with visual instruction tuning. In *CVPR*.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. Visual instruction tuning. In *NeurIPS*.

Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, et al. 2024b. Mmbench: Is your multi-modal model an all-around player? In *ECCV*.

OpenAI. 2024. Gpt-4o.

Jonas Pfeiffer, Gregor Geigle, Aishwarya Kamath, Jan-Martin O Steitz, Stefan Roth, Ivan Vulić, and Iryna Gurevych. 2021. xgqa: Cross-lingual visual question answering. *arXiv preprint arXiv:2109.06082*.

SakanaAI. 2024a. Ja-multi-image-vqa. https://huggingface.co/datasets/SakanaAI/JA-Multi-Image-VQA.

SakanaAI. 2024b. Ja-vg-vqa-500. https://huggingface.co/datasets/SakanaAI/JA-VG-VQA-500.

SakanaAI. 2024c. Ja-vlm-bench-in-the-wild. https://huggingface.co/datasets/SakanaAI/JA-VLM-Bench-In-the-Wild.

Turing. 2024. Llava-bench-in-the-wild (japanese). https://github.com/turingmotors/heron/tree/main/playground/data/llava-bench-in-the-wild.

Le Xue, Manli Shu, Anas Awadalla, Jun Wang, An Yan, Senthil Purushwalkam, Honglu Zhou, Viraj Prabhu, Yutong Dai, Michael S Ryoo, et al. 2024. xgen-mm (blip-3): A family of open large multimodal models. *arXiv preprint arXiv:2408.08872*.

Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. 2024a. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *CVPR*.

Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Botao Yu, Ge Zhang, Huan Sun, Yu Su, Wenhu Chen, and Graham Neubig. 2024b. Mmmu-pro: A more robust multi-discipline multimodal understanding benchmark. *arXiv preprint arXiv:2409.02813*.

Ge Zhang, Xinrun Du, Bei Chen, Yiming Liang, Tongxu Luo, Tianyu Zheng, Kang Zhu, Yuyang Cheng, Chunpu Xu, Shuyue Guo, et al. 2024. Cmmmu: A chinese massive multi-discipline multimodal understanding benchmark. *arXiv preprint arXiv:2401.11944*.

Textual Misunderstanding

Question: <image 1>の作者の代表作ではないものを選びなさい
(Select the work that is not a representative work by the author of <image 1>)
Option:
A. 乞食と女 (The Beggar and the Woman)
B. 大島の水汲み女 (Woman Fetching Water in Oshima)
C. 尿する裸僧 (Nude Monk Urinating)
D. 髪梳ける女 (Woman Combing Her Hair)

GPT-4o's Answer: この画像は「尿する裸僧」という作品です。したがって、選択肢の中でこの作者の代表作ではないものは「C. 尿する裸僧」です。
(This image is the artwork titled "Nude Monk Urinating." Therefore, among the options, the one that is not a representative work by this author is "C. 尿する裸僧.)

Ground Truth: D. 髪梳ける女

Error Reason:
GPT-4o correctly provides the title of <image 1>, but it fails to understand the question. Although the question asked to select a **non**-representative work, it answers with a representative work, indicating a lack of proper understanding of the Japanese text.

Image Recognition Error

Question: <image 1>で見られることで有名な動物は何でしょう？
(What animal is famous for being seen in the image?)
Option:
A. ヤンバルクイナ (Yanbaru rail)
B. イリオモテヤマネコ (Iriomote cat)
C. イヌワシ (Golden eagle)
D. トキ (Japanese crested ibis)

GPT-4o's Answer: この画像は日本の石垣島を示しています。石垣島で見られることで有名な動物は「イリオモテヤマネコ」です。したがって、正解はB. イリオモテヤマネコです。
(This image shows Ishigaki Island in Japan. The animal famous for being seen on Ishigaki Island is the 'Iriomote Cat'. Therefore, the correct answer is B. Iriomote Cat.)

Ground Truth: D. トキ

Error Reason:
GPT-4o mistakes Sado Island shown in <image 1> for Ishigaki Island which is another island in Japan.
(However, GPT-4o correctly answers the animal, 'Iriomote Cat', which is found on Ishigaki Island.)

Lack of Knowledge

Question: <image 1>には棟飾が取り付けられていますが、これを制作したとされる人物は誰ですか？
(An ornamental ridgepole is attached to <image 1>; who is the person said to have created it?)
Option:
A. 小橋川永昌 (Eisho Kobashigawa)
B. 平田典通 (Norimichi Hirata)
C. 河井寛次 (Kanji Kawai)
D. 金城次郎 (Jiro Kinjo)

GPT-4o's Answer: この画像は首里城の正殿を示しています。首里城の棟飾（龍頭）は、金城次郎（きんじょう じろう）によって制作されたとされています。したがって、正解はD. 金城次郎です。
(This image depicts the main hall of Shuri Castle. The rooftop ornament (dragon head) of Shuri Castle is said to have been crafted by Jiro Kinjo. Therefore, the correct answer is D. Jiro Kinjo.)

Ground Truth: B.平田典通

Error Reason:
GPT-4o correctly answers the name of the building in <image 1>, but it hallucinates when providing more detailed information about it.

Answer Rejection

Question: <image 1>の人物が研究していた病気として代表的なものは何でしょう？
(What is a representative disease that the person in <image 1> was researching?)
Option:
A. 黄熱病 (Yellow fever)
B. 結核 (Tuberculosis)
C. 赤痢 (Dysentery)
D. コレラ (Cholera)

GPT-4o's Answer: この画像の人物が誰かを特定することはできませんが、選択肢の中で黄熱病、結核、赤痢、コレラのいずれかを研究していた可能性があります。具体的な人物名がわかれば、その人物の研究分野を特定することができます。
(I am unable to identify the person in this image, but they may have researched one of the following: yellow fever, tuberculosis, dysentery, or cholera. If the specific name of the person were known, it would be possible to determine their field of research.)

Ground Truth: A. 黄熱病

Error Reason:
The person in the image is a famous historical figure in Japan, Hideyo Noguchi, but GPT-4o says it is unable to identify the person and refused to provide an answer.

Figure 6: Representative error types across models.

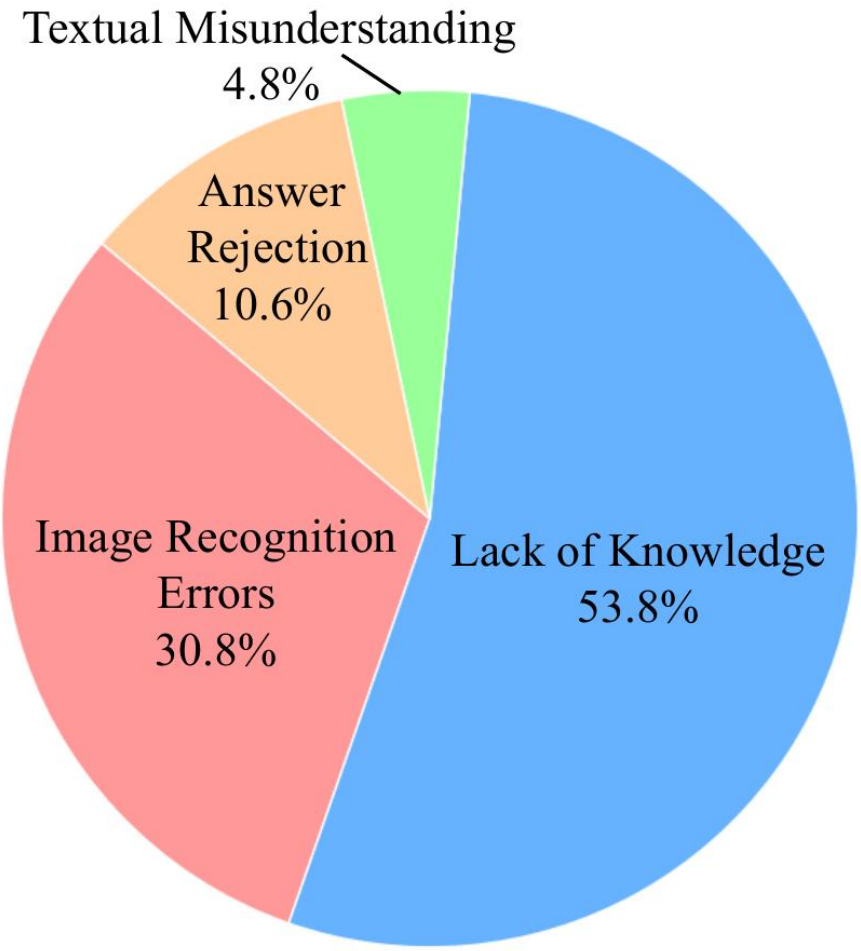

Figure 7: Error distribution over culture-specific subjects.